Development and Validation of a Hydraulic-Pressure Based Load Estimator for Static Wheel Loader

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1. Introduction

1.1 Background

In modern industrial facilities, autonomous wheel loaders are critical for managing bulk biomass, such as scooping wood piles for furnace feeding and pile organization. However, a major gap exists in their ability to physically perceive and quantify payloads during the excavation phase. Operating "blindly" without human senses, these loaders struggle with the variable density of wood piles, often resulting in empty or under-loaded buckets that lead to wasted fuel, unnecessary machine wear, and operational bottlenecks.

To resolve this, developing a Hydraulic-Pressure Based Load Estimator utilizing sensors on the lift cylinder will allow the system to calculate payload weight within a few seconds. This enables crucial payload verification, acting as a quality-control checkpoint that aborts transport and triggers a re-scoop if the load falls below a specific threshold, thereby eliminating inefficient "dry runs". Furthermore, knowing the precise weight allows the control system to implement adaptive kinematic control, dynamically adjusting velocity and acceleration profiles to account for changes in the machine's center of gravity and momentum. Transforming the loader into an intelligent, physically-aware system.

1.2 Scope and Operational Constraints

To ensure accurate payload verification and simplify the initial physical models, the development and validation of the load estimation system will be bound by the following scope and assumptions:

1. **Target Machinery**: The algorithms, physical dimensions, and sensor calibrations will be exclusively developed for the XCMG958 wheel loader.

2. **Static Conditions**: Rather than performing dynamic load weighing on the move, the mass estimation will be executed entirely under static conditions. The system will only record and process pressure sensor data after the boom has been lowered and the machine has come to a complete stop.

3. **Environmental and Posture Requirements**: To eliminate the effects of gravity offsets and external momentum, the loader must be stationary on flat, level ground during the measurement. Furthermore, the implement must be curled into a maximum tilt bucket posture to standardize the mechanical leverage.

4. **Center of Gravity Assumption**: For the physical model calculations, the payload is assumed to be distributed symmetrically, with the resultant center of gravity acting directly downward through the geometric center of the bucket.

5. **Boom Angular Range**: The system will only validate and process load estimations when the boom angle—tracked by the encoder at the boom-chassis pin—falls within a carefully specified operating range. Readings outside this angular window will be discarded to prevent kinematic calculation errors.

1.3 Objectives

The primary goal of this project is to develop and validate a hydraulic-pressure based load estimator specifically for the XCMG958 autonomous wheel loader. To achieve this, the project targets the following specific objectives:

1. **Preventing "Run Dry" Operations**: To establish a quality-control checkpoint immediately following the scooping phase. By verifying the payload, the system can abort transport and command a re-scoop if the load falls below a predefined weight threshold, thereby eliminating unnecessary empty transport cycles.

2. **High-Speed Estimation**: To ensure the real-time mass calculation is highly efficient, with the entire estimation process completed within a strict maximum timeframe of 3 seconds.

3. **Target Accuracy and Sensor Fusion**: To accurately calculate payload weight by fusing data from a pressure sensor installed on the lift cylinder with positional data from a rotary encoder at the boom-chassis pin. This integrated system must achieve an estimation accuracy of within $\pm 5\%$ of the machine's full load capacity (3.2 tons).

2. Modeling and Algorithm Development

The final load estimation is the result of a multi-stage process that filters raw sensor data, models physical behavior, and applies data-driven compensations to correct for model inaccuracies.

2.1 Simple Beam Model

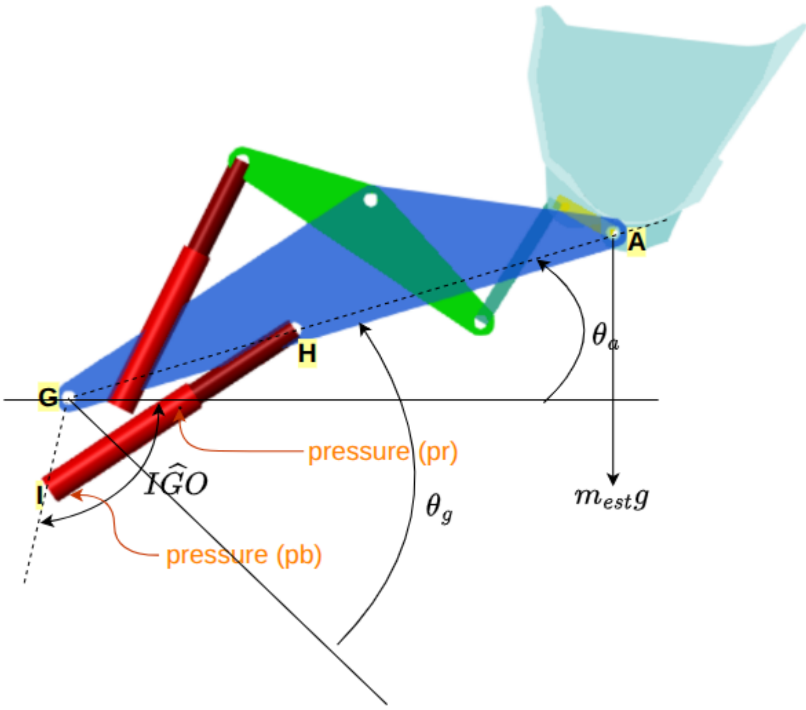

Figure 1: Simple Beam Model representing the loader's arm-bucket assembly

To translate the force from the hydraulic cylinders into an estimated load, the arm-bucket assembly is modeled as a **simple beam system**. This provides a physics-based initial estimate of the bucket load (w_{simple}) based on the known geometry and the calculated cylinder force.

1. **Cylinder Force (F_c)**:

$$F_c = 2 \cdot (A_b \cdot (p_b - p_{b\_offset}) - A_r \cdot (p_r - p_{r\_offset})) \cdot C \tag{1}$$

2. **Geometry Factor (a)**:

$$a = \sin(HIO) - \cos(HIO) \cdot \tan(\theta_a) \tag{2}$$

where HIO, GIH, and L_{ih} are intermediate geometric angles and lengths calculated from the loader's linkage constants and the current boom angle θ_a.

3. **Simple Load (w_{simple}):**

$$w_{simple} = \left(\frac{F_c \cdot L_{gh}}{L_{ag}} \right) \cdot \frac{a}{g} \tag{3}$$

Table 1: Parameters for Simple Beam Model

Parameter	**Value**	**Description**
A_b	0.02 m^2	Cross-sectional area of the cylinder bottom.
A_r	0.014 m^2	Cross-sectional area of the cylinder rod side.
C	1220.7 Pa/ADC	Conversion factor from sensor ADC to Pascals.
L_{gh}	1.630 m	Length of the main boom link.
L_{ag}	3.9294 m	Length of the lever arm from pivot to bucket.
L_{gi}	0.654 m	Length of the cylinder link.
IGO	1.786 rad	Fixed angle in the linkage geometry.
θ_a	$\theta_g - 0.62$ rad	Processed boom angle, corrected with an offset.
g	9.807 m/s^2	Acceleration due to gravity.

2.2 Load Estimation Algorithm

2.2.1 Signal Processing

Raw data from the hydraulic pressure sensors and the boom angle encoder is inherently noisy. To ensure a stable and reliable input for the downstream calculations, **Exponential Moving Average (EMA)** filters are applied to both sensor streams.

The filter is defined by the equation:

$$y_t = \alpha \cdot x_t + (1 - \alpha) \cdot y_{t-1} \tag{4}$$

Table 2: Signal Processing Parameters

Parameter	**Value**	**Description**
y_t	-	The filtered output at the current step.
x_t	-	The raw sensor input at the current step.
y_{t-1}	-	The filtered output from the previous step.
$\alpha_{pressure}$	0.001	Smoothing factor for pressure sensors (low cutoff).
α_{angle}	0.1	Smoothing factor for the angle encoder (higher cutoff).

2.2.2 Pressure Prediction

To provide a rapid estimation without waiting for the hydraulic system to fully stabilize, a pressure estimator is used. This model treats the pressure response during a downward boom movement as a **first-order system**. By analyzing the initial transient pressure change $p(t)$ after the boom stops, the model can quickly predict the final pressure value p_{final}.

The final pressure is estimated using the formula:

$$p_{final} = \frac{p(t) - p_{initial} \cdot e^{-t/\tau}}{1 - e^{-t/\tau}} \tag{5}$$

Table 3: Pressure Prediction Parameters

Parameter	Value	Description
p_{final}	-	The predicted final pressure.
$p(t)$	-	The filtered pressure measured at time t.
$p_{initial}$	-	Initial pressure when the boom stops ($t = 0$).
t	0 to 1.0 s	Time elapsed since the boom became static.
τ	15.0 s	Time constant of the first-order system ($0.25 \cdot T_s$).
T_s	60.0 s	Settling time of the hydraulic system.

2.2.3 Regression Model

Note: To get constant values ($c_1, c_2, c_3, k_1, k_2, k_3$), see the Calibration Procedure section. Constants must be adapted if a different wheel loader model is used.

Empty Bucket Pressure Offset: The weight of the loader's arm and bucket assembly exerts force on the hydraulic cylinders, resulting in a baseline pressure reading even when empty. This offset varies with the boom angle θ_g.

$$p_{b\_offset} = c_1 \cdot \theta_g + c_2 \tag{6}$$

$$p_{r\_offset} = c_3 \tag{7}$$

Table 4: Pressure Offset Model Parameters

Parameter	Value	Description
c_1	1699	Coefficient for the boom angle in p_{b_offset} model.
c_2	4023	Intercept for the p_{b_offset} model (ADC value).
c_3	413.3	Constant value for the p_{r_offset} model (ADC value).
θ_g	-	The boom angle from the encoder (rad).

Surface Fit Compensation: The simple beam model contains inaccuracies due to simplifying assumptions. A final compensation layer is applied via a linear model fitted to empirical datasets.

$$load_{final}\ [\text{ton}] = k_1 + k_2 \cdot w_{simple}\ [\text{kg}] + k_3 \cdot \theta_g\ [\text{rad}] \tag{8}$$

Table 5: Surface Fit Compensation Parameters

Parameter	**Value**	**Description**
w_{simple}	-	Initial load estimate from Simple Beam Model (kg).
θ_g	-	Raw boom angle from the encoder (rad).
k_1	-0.2494	Intercept term (ton).
k_2	1.0643×10^{-3}	Coefficient for the simple model's output.
k_3	0.51622	Coefficient for the boom angle.

3. Implementation

3.1 Software Architecture

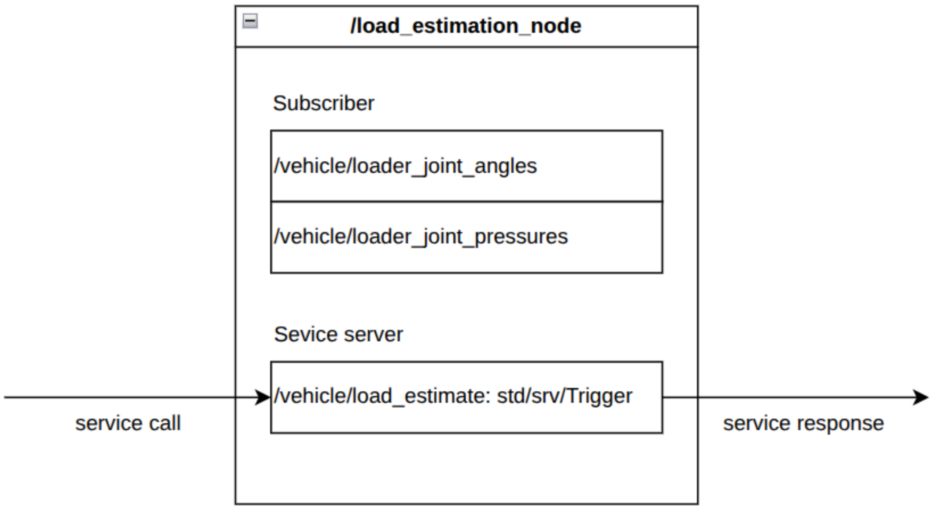

Figure 2: ROS 2 Structure for /load_estimation_node

The logic and multi-threaded architecture of the `load_estimate.py` ROS 2 node are separated into three modular flows: Continuous Callbacks, Service Initialization, and Data Collection.

3.1.1 Continuous Topic Callbacks (Angles & Pressures)

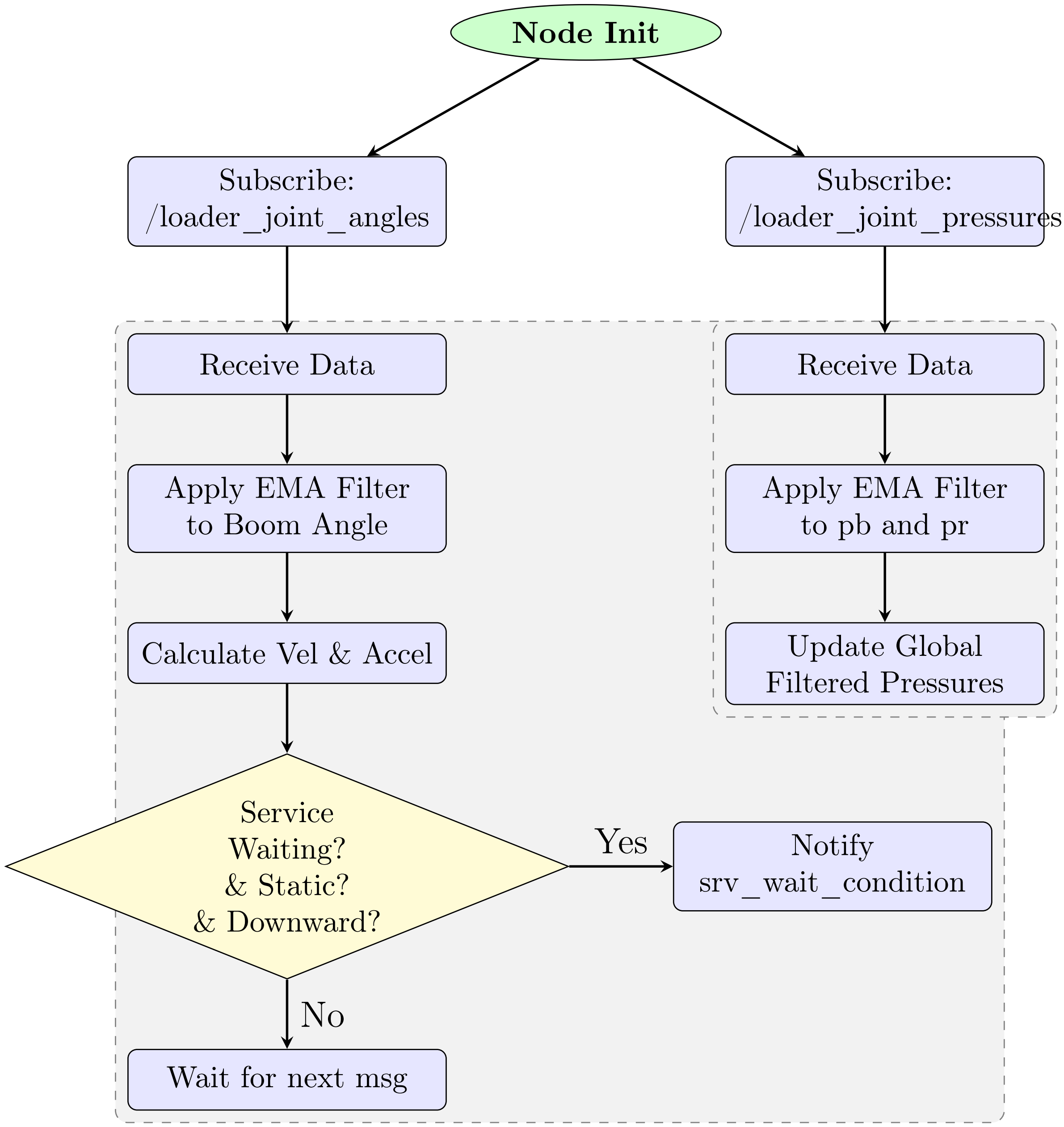

Figure 3: Continuous Topic Callbacks (Angles & Pressures Threading)

3.1.2 Service Call: Initialization & Wait Conditions

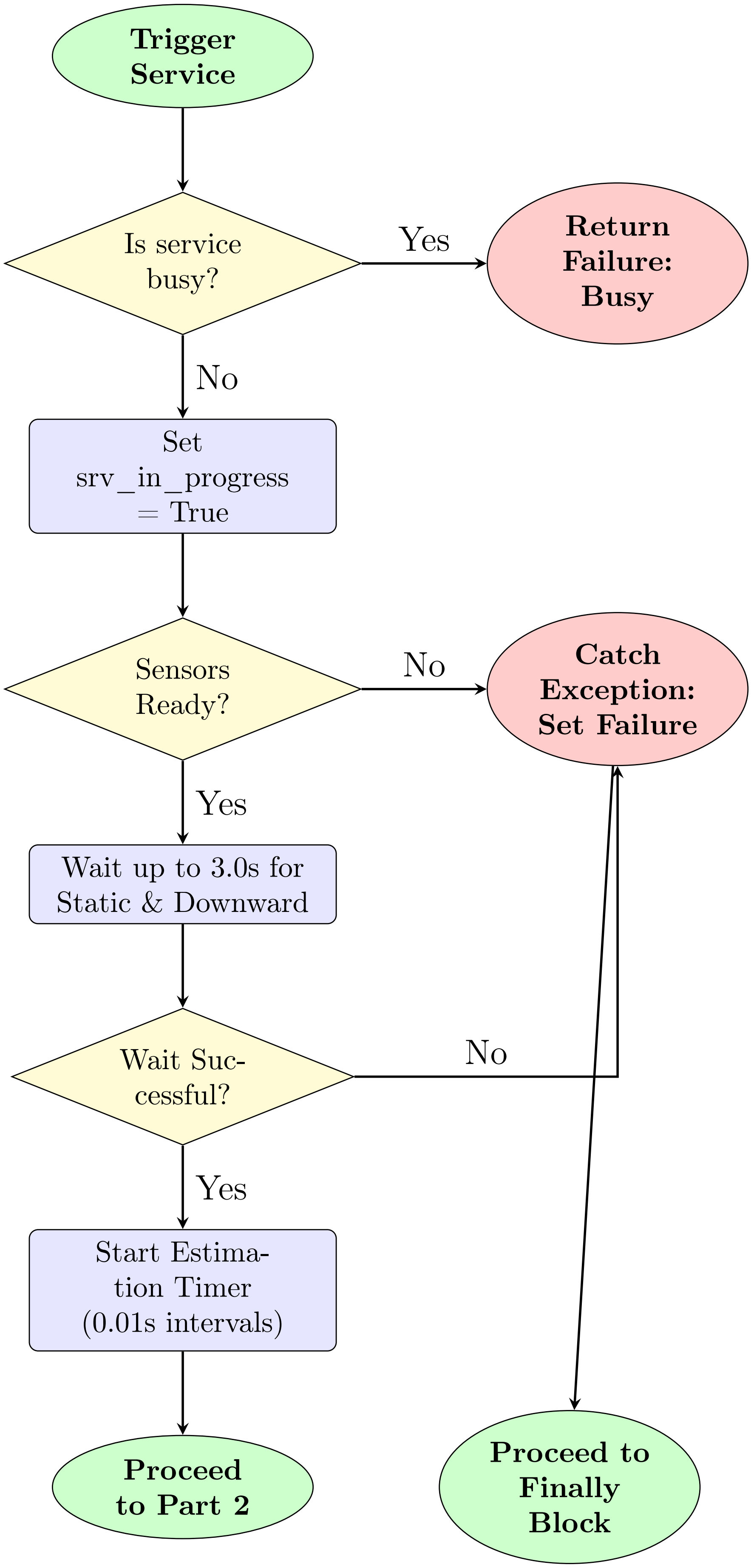

Figure 4: Service Call: Initialization & Wait Conditions

3.1.3 Service Call: Data Collection & Final Calculation

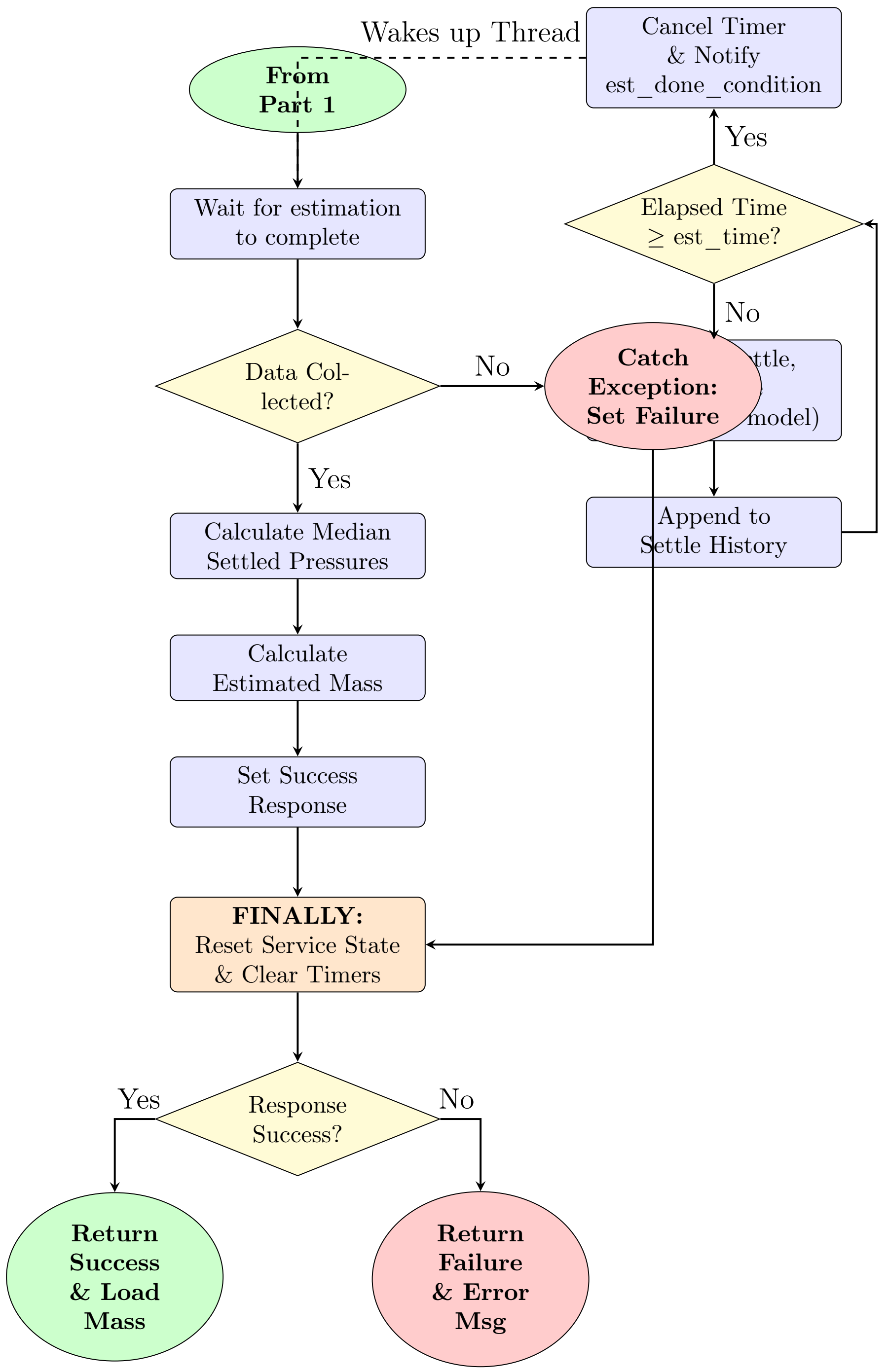

Figure 5: Service Call: Data Collection & Final Calculation

3.2 Hardware Component

- **PT5400 Electronic Pressure Sensor** (installed at lift cylinder).

4. Calibration Procedure

Note: This procedure must be executed if another wheel loader model is used.

4.1 Pressure Offset Model

Table 6: Empty Bucket Pressure Data for Offset Calibration

θ_g **(rad)**	$\mathbf{p_{bf}}$ **(ADC)**	$\mathbf{p_{rf}}$ **(ADC)**
0.1		
0.2		
0.3		
0.4		
0.5		
0.6		
0.7		

Fill the data in Table 1 while there is no load in the bucket. Use `linear` regression to find $p_{rf}(\theta_g)$, and treat p_{rf} as a constant (average value), where p_{rf} is the predicted pressure sensor value in ADC.

4.2 Surface Fit Compensation Model

Table 7: Surface Fit Compensate Calibration Data

θ_g **(rad)**	**3% Load**	**50% Load**	**100% Load**
0.1			
0.2			
0.3			
0.4			
0.5			
0.6			
0.7			

Fill the load values (derived from the Simple Beam Model) in Table 2, and utilize the `sklearn` library to find the regression function $load_{est}(w, \theta_g)$, where $load_{est}$ is the final estimated load in tons.

5. Results and Analysis

Note: These results utilize regression constants derived from 0.1, 1.66, and 3.22 tons calibration loads (using water in the bucket).

By varying the load (water) and boom angle, the following metrics were calculated:

- **RMSE_s**: Load root mean square error due to θ_g, indicating error affected by the boom angle.
- **RMSE_w**: Load root mean square error due to the payload, indicating error affected by the actual load in the bucket.

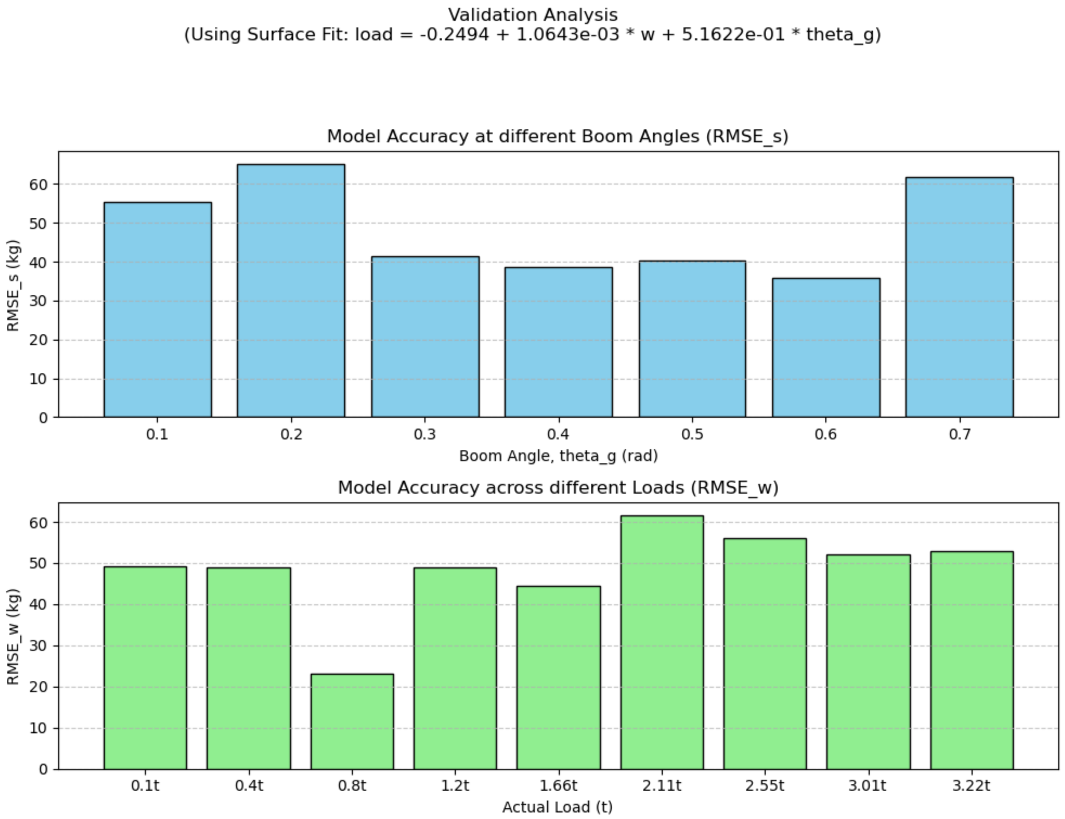

Figure 6: Validation Analysis across different boom angles and actual payloads.

The implemented methodology yields a load estimation with an accuracy of $\sim \pm 100$ **kg** for a full bucket capacity of 3 tons, under the specified operating conditions ($\theta_g = 0.3 - 0.6$ rad).

6. Conclusion

This project successfully developed and validated a Hydraulic-Pressure Based Load Estimator for the XCMG958 autonomous wheel loader. By fusing data from a lift cylinder pressure sensor with a boom-chassis pin rotary encoder, filtering raw signals using Exponential Moving Average (EMA) models, and processing the inputs through a physical Simple Beam Model coupled with a data-driven Surface Fit Compensation regression, the system delivers precise payload feedback.

Validated under static conditions on flat ground with the bucket at maximum tilt across a boom angle range of $\theta_g = 0.3$ to 0.6 rad, the estimator achieves an exceptional accuracy of $\sim \pm 100$ kg for a full bucket, easily satisfying the required $\pm 5\%$ (± 160 kg) accuracy threshold of the 3.2-ton full capacity.

Furthermore, by utilizing a first-order pressure prediction model that estimates settled hydraulic pressures using only 1.0 second of static data, the system bypasses the physical 60-second stabilization delay to complete the entire process well within the target 3-second operational window. This high-speed performance effectively prevents inefficient "run dry" transport cycles by enabling instant payload verification and re-scooping commands, while simultaneously providing the loader's central controller with the real-time physical feedback required for adaptive kinematic control to optimize velocity, acceleration, and driving safety.